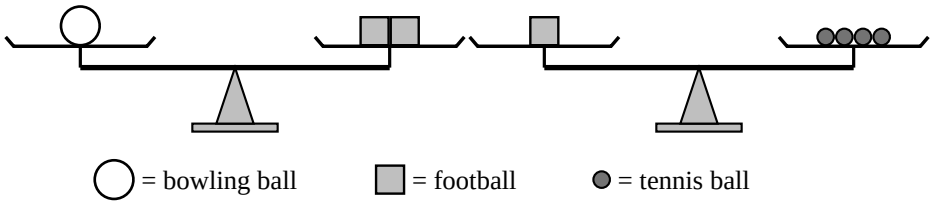
Math Kangaroo Practice

Grade 3-4 · 20 questions · harder · seed 49630

Part B: Each question is worth 4 points

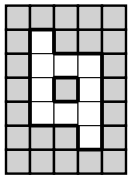
1. The two balances are level. One bowling ball weighs as much as 2 footballs, and one football weighs as much as 4 tennis balls.

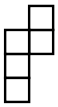


How many tennis balls weigh as much as 3 bowling balls?

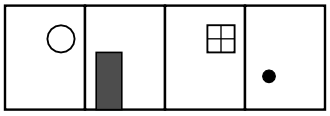
A. 30 ***B.*** 27 ***C.*** 25 ***D.*** 24 ***E.*** 9

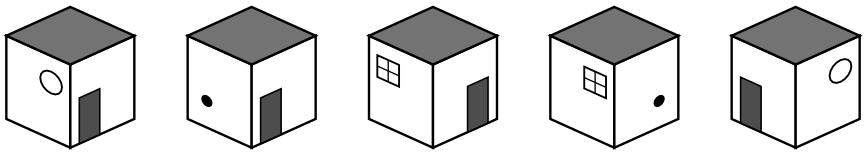
2. Two pieces that are exactly the same cover the empty area. Which piece is it? The pieces may be turned and flipped over.



A. ***B.*** ***C.*** ***D.*** ***E.*** 

3. The strip below is cut out and folded into the four walls of a house, with the pictures on the outside. A flat roof is put on top.





A B C D E

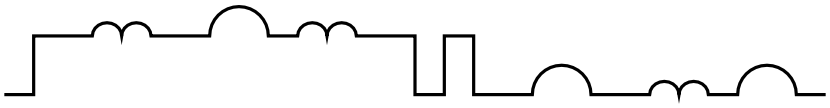
How many of the houses A–E can be made from the strip?

A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 4

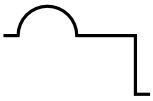
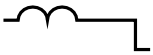
4. The odometer of a car shows 145543 km. Not all of its digits are different.
How many kilometres ago did the odometer last show a number with all different digits?

A. 145 ***B.*** 144 ***C.*** 135 ***D.*** 59 ***E.*** 1

5. Lena cut a strip of paper along this line.



Which piece **can** come from the cut line?

A.  ***B.*** ***C.*** ***D.***  ***E.***

6. Tom builds a brick 4 cubes long and 3 cubes tall from red and blue cubes. The outside is completely red and the inside is completely blue. Tom used exactly 8 blue cubes.
How many cubes deep is the brick?

A. 8 ***B.*** 7 ***C.*** 6 ***D.*** 5 ***E.*** 4

7. Ivan, Julien, Ben, Aisha and Leo were born in different months: April, December, October, January and March. Ben was born in a month that doesn't start with J. Ivan was born in spring. Aisha was born in spring. Julien was born in autumn. Ivan was not born in April.
Which statement is true?

A. Julien was born in March ***B.*** Ivan was born in December ***C.*** Ivan was born in January
D. Leo was born in January ***E.*** Aisha was born in December

8. A bar code is made of 27 black and white bars taking turns, with a black bar at each end. Each black bar is wide or narrow. There are 7 more white bars than wide black bars.

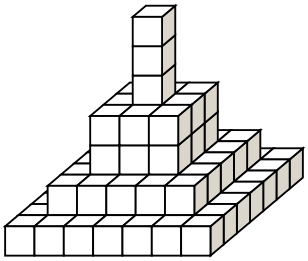
How many wide black bars are there?

A. 20 ***B.*** 7 ***C.*** 6 ***D.*** 5 ***E.*** 4

9. Five children, Grace, Hugo, David, Emma and Anna, sit on a bench. Grace always sits at the left end and Hugo always sits at the right end.
In how many different ways can the children sit on the bench?

A. 24 ***B.*** 6 ***C.*** 5 ***D.*** 4 ***E.*** 3

10. Sara built this pyramid out of identical cubes. Every layer is a full square, with no holes hidden inside.



How many cubes are in the pyramid?

A. 21 ***B.*** 63 ***C.*** 84 ***D.*** 95 ***E.*** 102

11. Tom reads a chapter from page 66 to page 111 (both included).
How many digits are printed in the page numbers of this chapter?

A. 225 ***B.*** 106 ***C.*** 104 ***D.*** 102 ***E.*** 90

Part C: Each question is worth 5 points

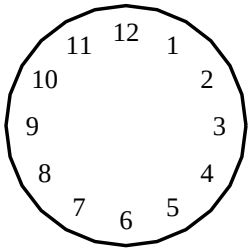
12. Mia has the cards 1, 2, 6 and 9. Using every card exactly once, with +, –, ×, ÷ and brackets, Mia wants to get the biggest possible number.
What is the biggest number Mia can get?

A. 194 ***B.*** 191 ***C.*** 162 ***D.*** 109 ***E.*** 107

13. Ana's new rug is 1 m 9 cm long. Four of these lengths are the same as that.
Which one is *not* the same?

A. 190 cm ***B.*** 1090 mm ***C.*** 1 m 90 mm ***D.*** 10 dm 9 cm ***E.*** 109 cm

14. A round cake with the numbers 1 to 12 around its edge is broken by one straight crack into two pieces. The sums of the numbers on the two pieces are equal.



Where does the crack go?

A. between 3 and 4, and between 9 and 10

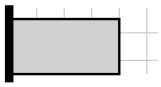
B. between 3 and 4, and between 8 and 9

C. between 2 and 3, and between 8 and 9

D. between 5 and 6, and between 8 and 9

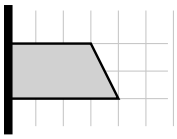
E. between 1 and 2, and between 9 and 10

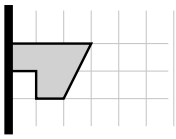
15. From a window, Tom sees a rectangular flag on a pole. The wind turns and folds the flag, but the side sewn to the pole never moves. Tom draws the shadow of the flag on the wall at five different moments. Flat, the flag is 4 squares wide and 2 squares tall (see the picture). Four of the drawings are wrong.

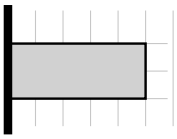


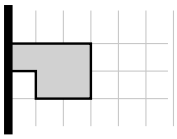
the flag, flat

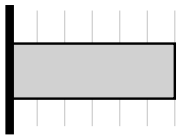
Which drawing can be right?

A. 

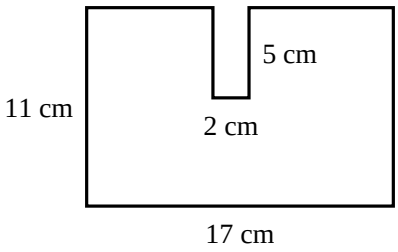
B. 

C. 

D. 

E. 

16. A 17 cm by 11 cm rectangle of cardboard has a rectangular notch 2 cm wide and 5 cm deep cut out of the middle of its top side.



What is the perimeter of the new shape?

A. 66 cm ***B.*** 64 cm ***C.*** 61 cm ***D.*** 56 cm ***E.*** 54 cm

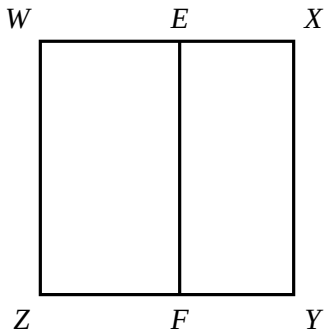
17. Tom, Hana, Leo, Julien and Zoe each have a different pet: a dog, a rabbit, a hamster, a goldfish and a parrot. Leo has a pet with two legs. Julien doesn't like hamsters. Hana has a pet with no legs at all. Tom has a pet that barks.
Which statement is **not** true?

A. Julien has a rabbit ***B.*** Tom has a dog ***C.*** Leo has a parrot
D. Zoe has a hamster ***E.*** Leo has a hamster

18. Finn's new rug is 2 m 8 cm long. Four of these lengths are the same as that.
Which one is *not* the same?

A. 28 cm ***B.*** 2080 mm ***C.*** 208 cm ***D.*** 2 m 80 mm ***E.*** 20 dm 8 cm

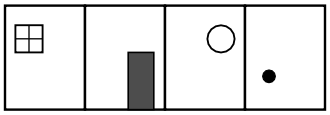
19. The square *WXYZ* is cut into two rectangles. The perimeter of *WEFZ* is 62 cm and the perimeter of *EXYF* is 58 cm.



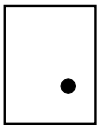
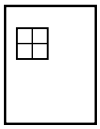
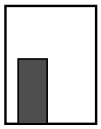
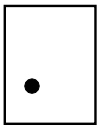
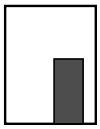
What is the perimeter of the square *WXYZ*?

A. 80 cm ***B.*** 60 cm ***C.*** 30 cm ***D.*** 20 cm ***E.*** 4 cm

20. The strip below is cut out and folded into the four walls of a house, with the pictures on the outside. A flat roof is put on top. You stand outside the house at a corner. The wall with the round window is on your left.



Which wall do you see on your right?

A.  ***B.***  ***C.***  ***D.***  ***E.*** 

Answer Key

1.	**D**	**2.**	**C**	**3.**	**B**	**4.**	**A**	**5.**	**B**	**6.**	**C**
7.	**D**	**8.**	**C**	**9.**	**B**	**10.**	**D**	**11.**	**C**	**12.**	**C**
13.	**A**	**14.**	**A**	**15.**	**A**	**16.**	**A**	**17.**	**E**	**18.**	**A**
19.	**A**	**20.**	**D**								

Solutions

1. D (24) — One bowling ball = 2 footballs = 2 × 4 = 8 tennis balls; 3 bowling balls = 3 × 8 = 24 tennis balls. [2002 Q9 · chain]

2. C — The empty area has 10 squares, so each piece has 5; every option has 5. Only two copies of piece C fit together into the shape of the hole. [2003 Q18 · two_copies]

3. B (1) — A house is right if its two walls are neighbours on the strip, in the same order, with the marks on the correct side. Right: A, so 1. [2003 Q12 · how_many_houses]

4. A (145) — Counting down from 145543, the first number with all different digits is 145398: 145543 – 145398 = 145. [2002 Q24 · last_time]

5. B — Only this piece is found in the line. The others don't fit — A: it has the right shapes but in the wrong order; C: its step is shorter than in the line; D: its step is shorter than in the line; E: its flat part between the two features has the wrong length. [2002 Q7 · can_belong]

6. C (6) — The blue inside is 2 long and 1 tall, so one layer of it has 2 × 1 = 2 cubes. 8 ÷ 2 = 4 layers of blue, plus the red front and back: 6. [2003 Q9 · how_deep]

7. D (Leo was born in January) — The only possible matching: Ivan – March; Julien – October; Ben – December; Aisha – April; Leo – January. [2002 Q18 · which_true]

8. C (6) — There are 14 black and 13 white bars. Wide: 13 – 7 = 6. [2003 Q17 · wide_bars]

9. B (6) — Only the 3 children in the middle can change places: 3 × 2 × 1 = 6 ways. [2002 Q22 · ends_fixed]

10. D (95) — Count layer by layer, from the bottom: 7 × 7 + 5 × 5 + 3 × 3 × 2 + 1 × 1 × 3 = 95. [2002 Q20 · count_pyramid]

11. C (104) — There are 46 pages: 34 × 2 + 12 × 3 = 104 digits. [2003 Q16 · from_page]

12. C (162) — (1 + 2) × 6 × 9 = 162. Multiplying by 1 changes nothing, so it is better to add the 1 to the smallest other card first. [2002 Q21 · biggest]

13. A (190 cm) — 1 m 9 cm = 109 cm = 1090 mm. But 190 cm = 190 cm. [2003 Q10 · not_equal]

14. A (between 3 and 4, and between 9 and 10) — Total 78, so each piece has 39: 4+5+6+7+8+9 = 39 and 10+11+12+1+2+3 = 39. [2002 Q11 · straight_crack]

15. A — Check two things: the pole edge must be covered all the way, and nothing may reach more than 4 squares from the pole. Only picture A passes both checks. [2002 Q15 · only_one_right]

16. A (66 cm) — The bottom of the notch (2 cm) replaces the 2 cm taken out of the top edge, and the two sides of the notch add 2 × 5 cm. Perimeter = 2 × (17 + 11) + 2 × 5 = 66 cm. [2002 Q6 · notch]

17. E (Leo has a hamster) — The only possible matching: Tom – dog; Hana – goldfish; Leo – parrot; Julien – rabbit; Zoe – hamster. [2002 Q18 · which_not_true]

18. A (28 cm) — 2 m 8 cm = 208 cm = 2080 mm. But 28 cm = 28 cm. [2003 Q10 · not_equal]

19. A (80 cm) — Together the two perimeters go around the square (4 sides) and along the cut twice (2 more sides): 62 + 58 = 120 cm = 6 sides. Side = 120 ÷ 6 = 20 cm, perimeter 4 × 20 = 80 cm. [2002 Q6 · two_perimeters]

20. D — Folded with the pictures outside, the strip keeps its order: on your right is the next wall on the strip. Its mark stays on the same side, so it is picture D. [2003 Q12 · right_neighbour]